worksheet05_group6

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Task 1

```
total_cancer <- sum(CRC_Incidence_Data$cancers)
total_population <- sum(CRC_Incidence_Data$cancers) + sum(CRC_Incidence_Data$noncancers)
overall_prob_cancer <- total_cancer / total_population
```

```
# logit_transform prob. so that we can fit linear model

# Fit the logistic regression model
logistic_model <- glm(cancerstatus ~ age, data = CRC_subset_expanded, family = "binomial")

summary(logistic_model)
```

```
##
## Call:
## glm(formula = cancerstatus ~ age, family = "binomial", data = CRC_subset_expanded)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -12.25366    1.80399  -6.793  1.1e-11 ***
## age           0.08851    0.03416   2.591  0.00958 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 5207.9  on 601916  degrees of freedom
## Residual deviance: 5201.1  on 601915  degrees of freedom
## AIC: 5205.1
##
## Number of Fisher Scoring iterations: 10
```

Age: 0.08851 coefficient is statistically significant since the p-value = 0.00958. So, the age has a statistically significant positive association with the log-odds of cancer. We can reject the null hypothesis the age and cancer are unrelated.

```
exp(0.08851)
```

```
## [1] 1.092545
```

For every increase of one year in age, the odds of having cancer is multiplied by 1.092545, which is a 9.25% increase.

```
# Fit the logistic regression model using cbind(cancers, noncancers) as the response

model_cancers_noncancers <- glm(
  cbind(cancers, noncancers) ~ age,
  data = CRC_subset,
  family = "binomial"
)

summary(model_cancers_noncancers)
```

```
##
## Call:
## glm(formula = cbind(cancers, noncancers) ~ age, family = "binomial",
##     data = CRC_subset)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -12.25366    1.80402  -6.792  1.1e-11 ***
## age           0.08851    0.03416   2.591  0.00958 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 10.6305  on 5  degrees of freedom
## Residual deviance:  3.8654  on 4  degrees of freedom
## AIC: 42.31
##
## Number of Fisher Scoring iterations: 4
```

Interpretation of coefficients is the same as above. **Null Deviance** represents the deviance of the null model (only intercept, no predictors). It shows how well the response is predicted by just its mean. A lower value indicates a better fit for null model. **Residual Deviance** is the deviance of the fitted model (age as predictor). It measures how well model fits data. A smaller residual deviance compared to null deviance ($3.86 < 10.63$) suggests that the predictor (age) explains some of the variability in the response (prob. of cancer). Both values are more easily interpretable compared to values above.

What's same?: Interpretation of intercept and age coefficients. Statistical significance measures. Model types (both logistic regression using family "binomial").

What's different?: How response data was presented. With cbind, groups are represented instead of individual observations. Moreover, deviance values and degrees of freedom are a difference. This is because of the input data structure. In this case we have much less rows (grouped observations instead of individual), which leads to less degrees of freedom. AIC value is also different, because it's derived from the deviances.

```
cat("Log-likelihood with intercept:", as.numeric(loglik_expanded), "\n")
```

```
## Log-likelihood with intercept: -2600.566
```

```r
cat("Log-likelihood without intercept:", as.numeric(loglik_expanded_no_intercept), "\n")
```

```
## Log-likelihood without intercept: -2624.185
```

```r
cat("Difference in log-likelihood (With - Without):", as.numeric(diff_loglik_expanded))
```

```
## Difference in log-likelihood (With - Without): 23.61902
```

The log-likelihood is a measure of how well the model fits data. A higher log-likelihood suggests a better fit. The difference in log-likelihood between the models with and without intercept reflects the contribution of intercept information to the fit.

The positive difference of 23.62 shows that adding the intercept term improves the model's fit to our data. So, the model with an intercept explains our data better.

```r
full_logistic_model <- glm(
  cbind(cancers, noncancers) ~ age + sex + race + registry + year_bin,
  data = CRC_Incidence_Data,
  family = "binomial"
)

summary(full_logistic_model)
```

```
##
## Call:
## glm(formula = cbind(cancers, noncancers) ~ age + sex + race +
##     registry + year_bin, family = "binomial", data = CRC_Incidence_Data)
##
## Coefficients:
##                       Estimate Std. Error   z value Pr(>|z|)
## (Intercept)         -1.199e+01  1.164e-02 -1030.283  < 2e-16 ***
## age                  8.720e-02  1.331e-04   655.346  < 2e-16 ***
## sexfemale           -3.330e-01  3.243e-03  -102.701  < 2e-16 ***
## raceblack            2.403e-01  5.812e-03    41.335  < 2e-16 ***
## raceother            2.199e-02  7.534e-03     2.919  0.00351 **
## registry02           1.551e-01  5.810e-03    26.697  < 2e-16 ***
## registry20           5.718e-02  5.778e-03     9.895  < 2e-16 ***
## registry21           1.219e-01  9.373e-03    13.006  < 2e-16 ***
## registry22           1.281e-01  6.041e-03    21.202  < 2e-16 ***
## registry23          -1.223e-01  8.215e-03   -14.881  < 2e-16 ***
## registry25          -1.489e-02  6.126e-03    -2.430  0.01510 *
## registry26          -1.752e-01  8.568e-03   -20.449  < 2e-16 ***
## registry27          -1.267e-02  7.413e-03    -1.709  0.08738 .
## year_bin(1975,1980]  1.373e-02  8.427e-03     1.629  0.10327
## year_bin(1980,1985]  4.203e-02  8.279e-03     5.077 3.84e-07 ***
## year_bin(1985,1990]  1.615e-02  8.214e-03     1.967  0.04924 *
## year_bin(1990,1995] -9.107e-02  8.257e-03   -11.029  < 2e-16 ***
## year_bin(1995,2000] -1.181e-01  8.215e-03   -14.375  < 2e-16 ***
## year_bin(2000,2005] -1.796e-01  8.228e-03   -21.827  < 2e-16 ***
## year_bin(2005,2010] -3.001e-01  8.291e-03   -36.194  < 2e-16 ***
## year_bin(2010,2015] -4.235e-01  9.170e-03   -46.188  < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 633128  on 18608  degrees of freedom
## Residual deviance:  39601  on 18588  degrees of freedom
## AIC: 110836
##
## Number of Fisher Scoring iterations: 4
```

The huge difference between Null Deviance and Residual Deviance (633128 vs. 39601) shows that extra information (age, sex, race, registry, year_bin) explains a very large amount of variability in cancer probability.

Each coefficient represents a log-odds change in cancer incidence compared to baseline group.

- age: Each one year increase in age increases odds of cancer by 1.0911149.
- sexfemale: Females have lower odds of cancer by `exp(-3.330e-01)` compared to males.
- raceblack: Black individuals have higher odds of cancer by 1.2716306 compared to white individuals.
- raceother: Other race individuals have higher odds of cancer by 1.0222336.
- year_bin: Effect of time period on cancer incidence compared to baseline (1970-1975). Coefficients get smaller over time, suggesting decreasing odds of cancer over time.

```
quad_age_model <- glm(cbind(cancers, noncancers) ~ age + I(age^2) + sex + race + registry + year_bin, fa

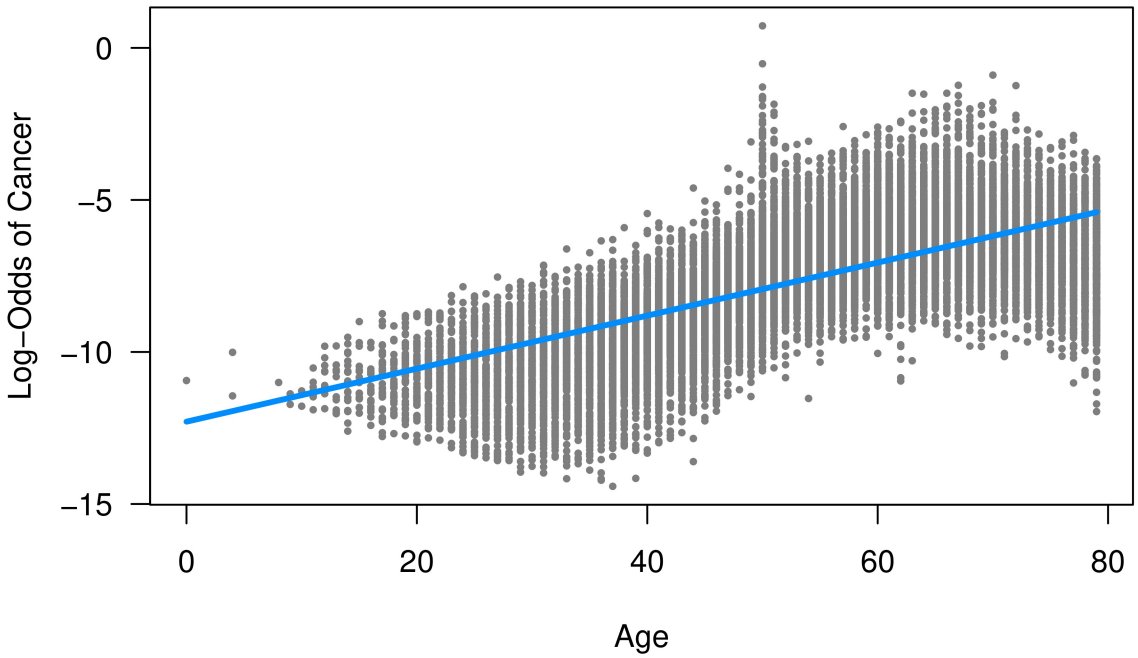
summary(quad_age_model)
```

```
##
## Call:
## glm(formula = cbind(cancers, noncancers) ~ age + I(age^2) + sex +
##     race + registry + year_bin, family = "binomial", data = CRC_Incidence_Data)
##
## Coefficients:
##                         Estimate Std. Error  z value Pr(>|z|)
## (Intercept)         -1.568e+01  3.942e-02 -397.891  < 2e-16 ***
## age                  2.178e-01  1.306e-03  166.734  < 2e-16 ***
## I(age^2)            -1.107e-03  1.086e-05 -101.944  < 2e-16 ***
## sexfemale           -3.239e-01  3.242e-03  -99.889  < 2e-16 ***
## raceblack            2.288e-01  5.807e-03   39.393  < 2e-16 ***
## raceother            1.314e-02  7.507e-03    1.750  0.08011 .
## registry02           1.521e-01  5.808e-03   26.190  < 2e-16 ***
## registry20           5.426e-02  5.776e-03    9.395  < 2e-16 ***
## registry21           1.161e-01  9.350e-03   12.413  < 2e-16 ***
## registry22           1.320e-01  6.040e-03   21.855  < 2e-16 ***
## registry23          -1.327e-01  8.215e-03  -16.156  < 2e-16 ***
## registry25          -1.632e-02  6.125e-03   -2.665  0.00771 **
## registry26          -1.731e-01  8.566e-03  -20.206  < 2e-16 ***
## registry27          -1.453e-02  7.406e-03   -1.962  0.04977 *
## year_bin(1975,1980]  2.021e-02  8.426e-03    2.398  0.01648 *
## year_bin(1980,1985]  5.364e-02  8.279e-03    6.479 9.24e-11 ***
## year_bin(1985,1990]  3.662e-02  8.215e-03    4.458 8.28e-06 ***
## year_bin(1990,1995] -6.380e-02  8.260e-03   -7.725 1.12e-14 ***
## year_bin(1995,2000] -8.921e-02  8.218e-03  -10.855  < 2e-16 ***
## year_bin(2000,2005] -1.582e-01  8.229e-03  -19.221  < 2e-16 ***
```

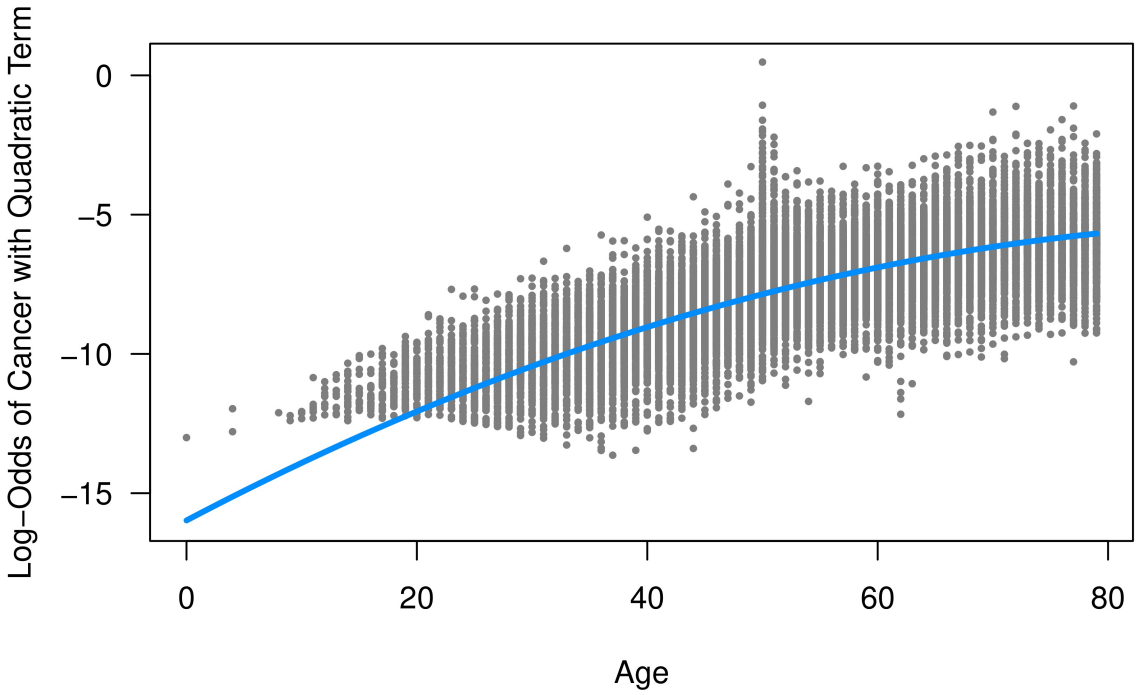
```
## year_bin(2005,2010] -2.935e-01  8.289e-03  -35.410  < 2e-16 ***
## year_bin(2010,2015] -4.272e-01  9.168e-03  -46.593  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 633128  on 18608  degrees of freedom
## Residual deviance:  27683  on 18587  degrees of freedom
## AIC: 98920
##
## Number of Fisher Scoring iterations: 5
```

Interpretation: Cancer risk increases with age until a certain age. After, it has almost no effect. This could be due to other causes of death outweighing cancer after a certain age.

```
library(visreg)
visreg(full_logistic_model, "age", ylab="Log-Odds of Cancer", xlab="Age")
```



```
visreg(quad_age_model, "age", ylab="Log-Odds of Cancer with Quadratic Term", xlab="Age")
```



The gray points are partial residuals: how each data point deviates from the expected log-odds. The residuals seem to decrease with the added quadratic term. The quadratic term allows the model to reflect the non-linear relationship between age and cancer.

In [22]:
```
import scanpy as sc
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
import time
```

Loading the Preprocessed PBMC Dataset

In [11]:
```
data = sc.datasets.pbmc3k_processed()
print(data)
```

```
AnnData object with n_obs × n_vars = 2638 × 1838
    obs: 'n_genes', 'percent_mito', 'n_counts', 'louvain'
    var: 'n_cells'
    uns: 'draw_graph', 'louvain', 'louvain_colors', 'neighbors', 'pca', 'rank_genes_groups'
    obsm: 'X_pca', 'X_tsne', 'X_umap', 'X_draw_graph_fr'
    varm: 'PCs'
    obsp: 'distances', 'connectivities'
```

In [19]:
```
print(data.obs)
```

```
                 n_genes  percent_mito  n_counts          louvain
AAACATACAACCAC-1      781      0.030178   2419.0      CD4 T cells
AAACATTGAGCTAC-1     1352      0.037037   4903.0        B cells
AAACATTGATCAGC-1     1131      0.008897   3147.0      CD4 T cells
AAACCGTGCTTCCG-1      960      0.017431   2639.0  CD14+ Monocytes
AAACCGTGTATGCG-1      522      0.012245    980.0       NK cells
...                   ...           ...      ...            ...
TTTCGAACTCTCAT-1     1155      0.021244   3459.0  CD14+ Monocytes
TTTCTACTGAGGCA-1     1227      0.009294   3043.0      CD4 T cells
TTTCTACTTCCTCG-1      622      0.021073   1004.0        B cells
TTTGCATGAGAGGC-1      454      0.020548   1622.0        B cells
TTTGCATGCTCAC-1       724      0.008065   1569.0      CD4 T cells

[2638 rows x 4 columns]
```

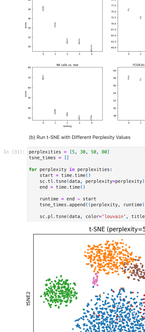
In [16]:
```
print(data.var)
```

```
        n_cells
index
TNFRSF4     155
CPSF3L      282
ATAD3C        9
C1orf86     501
MTRR1       688
...         ...
ICOSLG       34
SUMO3       578
SLC35A1      31
S100B        94
PRMT2       588

[1838 rows x 1 columns]
```

(a) Perform PCA as a first dimensionality reduction step and plot your results

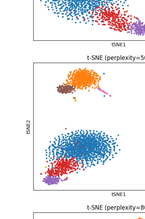
In [17]:
```
sc.tl.pca(data, svd_solver='arpack')
sc.pl.pca(data, color='louvain', title="PCA: Louvain Clusters")
```



Louvain Clustering: The plot overlays Louvain clusters to ensure group similarity.

1. Clear separation of clusters like B cells (green) and CD14+ Monocytes (orange), and NK cells indicating distinct transcriptional profiles.
2. CD4 and CD8 T cells (blue and red) are closely located, reflecting shared lineage and divergence or overlapping expression profiles, as they are types of T lymphocytes in immune system.
3. Some mixing or overlap exists, indicates transitional states or technical noise.

In [34]:
```
#top marker genes per cluster(cell type)
sc.tl.rank_genes_groups(data, 'louvain', method='t-test')
sc.pl.rank_genes_groups(data, n_genes=5, sharey=False)
```



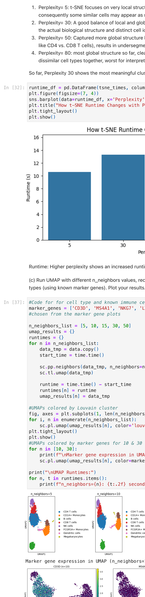
(b) Run t-SNE with Different Perplexity Values

In [31]:
```
perplexities = [5, 30, 50, 80]
time_times = []

for perplexity in perplexities:
    start = time.time()
    sc.tl.tsne(data, perplexity=perplexity)
    end = time.time()

    runtime = end - start
    time_times.append(perplexity, runtime)

    sc.pl.tsne(data, color='louvain', title=f"t-SNE (perplexity={perplexity})")
```

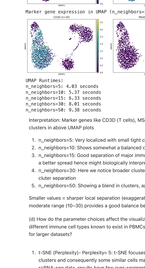


Interpretation:

1. Perplexity: 5 : t-SNE focuses on very local structure,and is sensitive to local noise results in over-segmented(overlapped) clusters and consequently, some similar cells may appear as multiple clusters.
2. Perplexity: 30 . A good balance of local and global structure in scRNA-seq data, results have few over-segmented clusters, ideally preserves the actual biological structures and distinct cell identities.
3. Perplexity: 50: Captured more global structure but chances are it oversimplified some clearly distinct clusters(i.e., between similar cell types like CD4 vs. CD8 T cells), results in undersegmentation.
4. Perplexity: 80: most global structure so far, clear under segmentation, multiple cell types groups together in a single cluster, clubbing dissimilar cell types together, worst for interpretation.

So far, Perplexity 30 shows the most meaningful clustering without distorting local or global relationships much

In [32]:
```
runtime_df = pd.DataFrame(time_times, columns=['Perplexity', 'Runtime (s)'])
plt.figure(figsize=(7, 4))
sns.barplot(data=runtime_df, x='Perplexity', y='Runtime (s)')
plt.title('How t-SNE Runtime Changes with Perplexity')
plt.tight_layout()
plt.show()
```



Runtime: Higher perplexity shows an increased runtime due to more complex optimization and further over-simplification of the data

(c) Run UMAP with different n_neighbors values, record the runtime for each run, and analyze how n_neighbors affects the visualization of cell types (using known marker genes). Plot your results.

In [33]:
```
#Code for for cell type and known immune cell marker genes
marker_genes = ['CD3D', 'MS4A1', 'NKG7', 'LYZ']  # T cells, B cells, NK cells, Monocytes
#chosen from the marker gene plots

n_neighbors_list = [5, 10, 15, 30, 50]
umap_results = {}
runtimes = []
for n in n_neighbors_list:
    data_tmp = data.copy()
    start_time = time.time()

    sc.pp.neighbors(data_tmp, n_neighbors=n, use_rep='X_pca')
    sc.tl.umap(data_tmp)

    runtime = time.time() - start_time
    runtimes.append(runtime)
    umap_results[n] = data_tmp

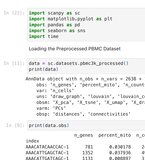
#UMAPs colored by Louvain cluster
fig, axes = plt.subplots(1, len(n_neighbors_list), figsize=(4 * len(n_neighbors_list), 4))
for i, n in enumerate(n_neighbors_list):
    sc.pl.umap(umap_results[n], color='louvain', title=f'n_neighbors={n}', show=False, ax=axes[i])
plt.tight_layout()
plt.show()
#UMAPs colored by marker genes for 10 & 30
for n in [10, 30]:
    print(f"\nMarker gene expression in UMAP (n_neighbors={n}):")
    sc.pl.umap(umap_results[n], color=marker_genes, title=[f"{gene} (n={n})" for gene in marker_genes])

print("\nUMAP Runtimes:")
for n, t in runtimes.items():
    print(f"n_neighbors={n}: {t:.2f} seconds")
```



Marker gene expression in UMAP (n_neighbors=10):

Marker gene expression in UMAP (n_neighbors=30):



UMAP Runtimes:
n_neighbors=5: 4.83 seconds
n_neighbors=10: 5.37 seconds
n_neighbors=15: 5.35 seconds
n_neighbors=30: 8.41 seconds
n_neighbors=50: 9.36 seconds

Interpretation: Marker genes like CD3D (T cells), MS4A1 (B cells), NKG7 (NK cells), and LYZ (Monocytes) help identify distinct immune cell clusters in above UMAP plots

1. n_neighbors=5: Very localized with small tight cluster, overfits to local structure; less generalizable
2. n_neighbors=10: Shows somewhat a balanced cluster, almost all immune cell types are separatly clustered
3. n_neighbors=15: Good separation of major immune types, similar to n=10, however dendritic and megakaryotes are closely clustered but with a better spread hence might biologically interpretable
4. n_neighbors=30: Here we notice broader clusters, smoother transitions, they are less tight as compared to previous n values, but still a good cluster separation
5. n_neighbors=50: Showing a blend in clusters, again can result in underfitting, but we can use this for lineage discovery or high-level trends

Smaller values = sharper local separation (exaggerated differences). Larger values = smooth transitions (hide fine-grained differences). A moderate range (10–30) provides a good balance between clear separation and meaningful biological relationships.

(d) How do the parameter choices affect the visualization of the cell populations? Which method better preserves the relationships between different immune cell types known to exist in PBMCs? How do computation times compare between t-SNE and UMAP, and why might this matter for larger datasets?

1. t-SNE (Perplexity): Perplexity- 5: t- SNE focuses on very local structure,and is sensitive to local noise results in over-segmented(overlapped) clusters and consequently some similar cells may appear as multiple clusters. Perplexity- 30: A good balance of local and global structure in scRNA-seq data, results have few over-segmented clusters, ideally preserves the actual biological structure and distinct cell identities. Perplexity- 50: Captured more global structure but chances are it oversimplified some clearly distinct clusters(i.e., between similar cell types like CD4 vs. CD8 T cells), results in undersegmentation. Perplexity- 80: most global structure so far, clear under segmentation, multiple cell types groups together in a single cluster, clubbing dissimilar cell types together, worst for interpretation.
2. UMAP (n_neighbors)-n_neighbors=5: Very localized with small tight cluster, overfits to local structure; less generalizable n_neighbors=10: Shows somewhat a balanced cluster, almost all immune cell types are separatly clustered n_neighbors=15: Good separation of major immune types, similar to n=10, however dendritic and megakaryotes are closely clustered but with a better spread hence might biologically interpretable n_neighbors=30: Here we notice broader clusters, smoother transitions, they are less tight as compared to previous n values, but still a good cluster separation n_neighbors=50: Showing a blend in clusters, again can result in underfitting, but we can use this for lineage discovery or high-level trends
3. UMAP typically preserves global structure better as the immune cell types like T cells, B cells, Monocytes, and NK cells maintain consistent spatial relationships across clusters. Hence it is good for exploring gradients, lineages, or transitions in monocytes or t cells. Whereas t-SNE is better for local detail. It is excellent in sense of tight clustering of well-defined populations (like B cells, NK cells, T cells etc) but they are poor at showing overall proximity or developmental relationships (e.g., CD4 and CD8 T cells may appear far apart even if biologically close). Therefore, t-SNE seems good for local cluster separation and UMAP for global biological relations.
4. Computation time of UMAP is much faster, and it is efficient on large datasets.

Task3

a- Clustering is an instance of unsupervised learning. --> True

b- In K-means clustering, we directly maximize the between-cluster-dissimilarity. --> False (minimize within cluster sum of squares)

c- In K-means clustering, we are guaranteed to find the global optimum if we re-run the clustering on the same data multiple times. --> False (converge to local minima)

d- In hierarchical clustering, the height of the cut in a dendrogram controls the number of clusters obtained. --> True

e- In hierarchical clustering, the lower in the tree a fusion between groups occurs, the more similar the elements of the groups are to each other. --> True

f- In a dendrogram with 10 leafs (observations), there are 128 equivalent reorderings of the dendrogram. --> False ($2^{n-1} = 2^9 = 512 != 128$)